

Year 7 • Lesson 10.08 Comparing data sets: answers (teacher copy)

10.08 Comparing data sets (booklet p. 3)

Warm-up

1. 14
2. 30
3. 10
4. spread out

WE DO examples

1. Ex 1: A mode 7, range 5; B modes 8 and 9, range 8
2. Ex 2: A median 7, B median 7.5. B did slightly better (higher median); A is more consistent (smaller range)
3. Ex 3: plot with stems 2–5, key e.g. 3 | 4 = 34
4. Ex 4: P median 40, range 24; Q median 32, range 17. P did better (higher median); Q is more consistent (smaller range)
5. Ex 5: Amy mean 13.33, range 3; Beth mean 14, range 17. Amy is more consistent
6. Ex 6: e.g. Amy: similar mean but far more reliable. (Accept Beth if argued: she can score very high.)

Set A

1. 12
2. 15
3. 18
4. 63
5. 25
6. 35
7. 33
8. 42
9. 9
10. Gym B (median 39 vs 32)

Set B

1. P
2. S
3. P
4. S
5. P
6. S
7. P
8. S
9. P
10. S

Set C

1. Stems 14–17; girls' leaves on the left
2. 175 cm, boys
3. girls 160, boys 158
4. girls 161, boys 158
5. girls 22, boys 30
6. girls 159.18, boys 158.5
7. The girls have a slightly higher mean and median, but the squads are very similar
8. The girls (smaller range)
9. girls 6, boys 4
10. e.g. The squads have similar average heights. The boys' heights are more spread out.
11. Primary
12. e.g. the girls: higher mean and median, and more consistent heights

Set D

1. Stems 2–9; 7R leaves on the left
2. 97, 7T
3. mean 67.47, median 68, mode 72, range 41
4. mean 70.67, median 74, mode no mode, range 74
5. 7T has a higher mean (70.67 vs 67.47) and median (74 vs 68)
6. 7R: range 41 vs 74
7. 23: it is well below the rest of 7T
8. mean 74.07, range 59
9. Primary: she collected the marks herself
10. e.g. 7T scored higher on average but their marks were much more spread out; 7R was more consistent

Extension

1. $96 + 204 = 300$; $300 \div 20 = 15$
2. The bigger group has more effect on the total
3. e.g. 5, 5, 5, 5, 5 and 1, 3, 5, 7, 9

Exit ticket

1. Level 1: 30, Team A
2. Level 2: A 22.5, B 26
3. Level 3: Team B (range 11 vs 18)